Rahul Parhi

curriculum vitæ

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Research Summary

My general interests are in **applied harmonic analysis**, **applied functional analysis**, and the **mathematics of data**, with connections to **signal processing**, **machine learning**, **statistics**, and **optimization**. Currently, my research focuses on the mathematical foundations of neural networks and deep learning, applied and computational aspects of functional and harmonic analysis, and applications in inverse problems and computational imaging.

Employment

2024– **Assistant Professor**, *University of California, San Diego*, La Jolla, California, USA
Department of Electrical and Computer Engineering

2022–2024 **Collaborateur Scientifique**, *École Polytechnique Fédérale de Lausanne*, Lausanne, Switzerland
Institut de Génie Électrique et Microtechnique

Summer 2018 **Software Development Engineer Intern**, *Amazon.com, Inc.*, Seattle, Washington, USA

Summer 2017 **Software Development Engineer Intern**, *Amazon.com, Inc.*, Seattle, Washington, USA

Education

2018–2022 **Ph.D., Electrical Engineering**, *University of Wisconsin–Madison*, Madison, Wisconsin, USA

2018–2019 **M.S., Electrical Engineering**, *University of Wisconsin–Madison*, Madison, Wisconsin, USA

2015–2018 **B.S., Computer Science**, *University of Minnesota, Twin Cities*, Minneapolis, Minnesota, USA

2015–2018 **B.S., Mathematics**, *University of Minnesota, Twin Cities*, Minneapolis, Minnesota, USA

Selected Honors and Awards

2026 Vasil A. Popov Prize
for *"distinguished research accomplishments in Approximation Theory and related areas of mathematics"*

2026 Hellman Fellowship, University of California
awarded $50,000 to support the project *"Principled Deep Learning for Imaging Inverse Problems"*

2026 SIAM Review SIGEST Award
for *"an outstanding paper of general interest"*

2025 NVIDIA Academic Grant Award
awarded 2× RTX PRO 6000 Blackwell Max-Q Workstation Edition GPUs

2024 Conference on Parsimony and Learning (CPAL) Rising Stars Award

2023 Harold Peterson Outstanding Dissertation Award, University of Wisconsin–Madison

2020 NSF Graduate Research Fellowship
for the project *"Bringing Approximation Theory to Data Science"*

2017 Astronaut Scholarship

2017 Barry M. Goldwater Scholarship

Publications & Preprints

See: `https://sparsity.ucsd.edu/files/rahulparhi-pubs.pdf`

Talks & Presentations

See: `https://sparsity.ucsd.edu/files/rahulparhi-talks.pdf`

Teaching

Teaching materials and evaluations are available online: `https://sparsity.ucsd.edu/teaching/`

University of California, San Diego

Spring 2027	ECE 172A: Introduction to Intelligent Systems (a.k.a. Image Processing)	*Instructor*
Winter 2027	ECE 285: Elements of Functional Analysis in Data Science	*Instructor*
Fall 2026	ECE 251C: Filter Banks and Wavelets	*Instructor*
Spring 2026	ECE 161C: Applications of Digital Signal Processing (a.k.a. Image Processing)	*Instructor*
Fall 2025	ECE 251C: Filter Banks and Wavelets	*Instructor*
Winter 2025	ECE 172A: Introduction to Intelligent Systems (a.k.a. Image Processing)	*Instructor*
Fall 2024	ECE 251C: Filter Banks and Wavelets	*Instructor*

École Polytechnique Fédérale de Lausanne

Spring 2024	MICRO 512: Image Processing II	*Teaching Assistant*
Fall 2023	MICRO 511: Image Processing I	*Teaching Assistant*
Spring 2023	MICRO 512: Image Processing II	*Teaching Assistant*
Fall 2022	MICRO 511: Image Processing I	*Teaching Assistant*

University of Wisconsin–Madison

Spring 2022	ECE/CS 761: Mathematical Foundations of Machine Learning	*Teaching Assistant*
Fall 2021	ECE/CS/ME 532: Matrix Methods in Machine Learning	*Teaching Assistant*
Spring 2021	ECE 901: Nonparametric Methods in Data Science	*Co-Instructor and Teaching Assistant*
Fall 2020	ECE/CS/ME 532: Matrix Methods in Machine Learning	*Teaching Assistant*
Spring 2020	ECE/CS 761: Mathematical Foundations of Machine Learning	*Teaching Assistant*

University of Minnesota, Twin Cities

Spring 2018	MATH 5651: Basic Theory of Probability and Statistics	*Teaching Assistant*
Fall 2017	CSCI 1933: Introduction to Algorithms and Data Structures	*Teaching Assistant*
Spring 2017	CSCI 1933H: Honors Introduction to Algorithms and Data Structures	*Teaching Assistant*
Fall 2016	CSCI 1933: Introduction to Algorithms and Data Structures	*Teaching Assistant*
Spring 2016	CSCI 1933: Introduction to Algorithms and Data Structures	*Teaching Assistant*

Research Advising and Mentoring

University of California, San Diego

Winter 2026– **Soroush Mesforush**, *Ph.D. Student*, Electrical and Computer Engineering

Spring 2025– **Henry Pritchard**, *B.S./M.S./Ph.D. Student*, Electrical and Computer Engineering

Winter 2025– **Anıl Kamber**, *Ph.D. Student*, Electrical and Computer Engineering

Winter 2025– **Tongtong Liang**, *Ph.D. Student*, Mathematics

Fall 2024– **Luke McDermott**, *Ph.D. Student*, Electrical and Computer Engineering

Professional Service

University Service at UCSD

2025– Organizing Committee for the Optimization for ML and AI Seminar

2024– Teaching Innovations and Undergraduate Affairs Committee, Department of Electrical and Computer Engineering

Meetings Organized and Chaired

November 2026 Organizer and chair (with Greg Ongie) for the session on "function-space perspectives on compositionality and depth in DNNs", SIAM Conference on Mathematics of Data Science (MDS), Salt Lake City, Utah, USA, 2026

July 2025 Organizer and chair (with Thibaut Horel) for the session on "relaxations of optimization problems and extreme point results in infinite dimensions", International Conference on Continuous Optimization (ICCOPT), Los Angeles, California, USA, 2025

July 2024 Chair for the session on "nonsmooth and hierarchical optimization in machine learning", International Symposium on Mathematical Programming (ISMP), Montréal, Québec, Canada, 2024

Reviewer for the Following Journals and Conferences

- Applied Mathematics Journals
  - Applied and Computational Harmonic Analysis
  - Constructive Approximation
  - IMA Journal of Applied Mathematics
  - IMA Journal of Numerical Analysis
  - Integral Transforms And Special Functions
  - Journal of Computational and Applied Mathematics
  - SIAM Journal on Imaging Sciences
  - SIAM Journal on Mathematical Analysis
  - SIAM Journal on Mathematics of Data Science
  - SIAM Journal on Scientific Computing
- Probability/Statistics Journals
  - Annals of Applied Probability
  - Annals of Statistics
  - Biometrika
- Machine Learning/Artificial Intelligence Journals and Conferences
  - IEEE Transactions on Neural Networks and Learning Systems
  - International Conference on Artificial Intelligence and Statistics (AISTATS)
  - International Conference on Machine Learning (ICML)
  - Journal of Machine Learning Research
  - Learning for Dynamics & Control Conference (L4DC)
  - Machine Learning and Systems (MLSys)
  - Neural Information Processing Systems (NeurIPS)
  - Neural Networks
  - Neurocomputing
- Signal Processing/Information Theory Journals and Conferences
  - IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
  - IEEE Open Journal of Signal Processing
  - IEEE Signal Processing Letters
  - IEEE Signal Processing Magazine
  - IEEE Transactions on Information Theory
  - IEEE Transactions on Signal Processing

Other Reviewing

2024– Regular contributor of Mathematical Reviews® for MathSciNet®

Professional Societies

2015, 2022– IEEE Member

2024– SIAM Member

Miscellaneous

Other Honors and Awards

2026 ICML Silver Reviewer Award

2022 IEEE Signal Processing Society Travel Grant

2022 University of Wisconsin–Madison ECE Dissertator Travel Award

2018 LUCID NSF Research Traineeship (NRT) from NSF grant 1545481 (2 years)

2018 NSF Graduate Research Fellowship Program Honorable Mention

2015 IEEE Circuits and Systems Society Travel Grant

2013–2015 Post-Secondary Enrollment Options (PSEO) Student at the University of Minnesota, Twin Cities
Enrolled full-time at the University of Minnesota–Twin Cities during 11th and 12th grades through PSEO.

2008–2013 University of Minnesota Talented Youth Math Program (UMTYMP)
Completed differential and integral calculus, linear algebra, differential equations, set theory, methods of proof, and multivariable calculus at the University of Minnesota–Twin Cities at the end of 10th grade through UMTYMP.

Erdős Number: **3** via many different paths, including

- Paul Erdős and Ambikeshwar Sharma. "On Tchebycheff Quadrature". In: *Canadian Journal of Mathematics* 17 (1965), pp. 652–658. DOI: 10.4153/CJM-1965-064-X
- Ronald Devore, Amram Meir, and Ambikeshwar Sharma. "Strongly and Weakly Non-Poised HB Interpolation Problems". In: *Canadian Journal of Mathematics* 25.5 (1973), pp. 1040–1050. DOI: 10.4153/CJM-1973-111-8
- Ronald DeVore, Robert D. Nowak, **Rahul Parhi**, and Jonathan W. Siegel. "Weighted Variation Spaces and Approximation by Shallow ReLU Networks". In: *Applied and Computational Harmonic Analysis* 74.101713 (2025), pp. 1–22. DOI: 10.1016/j.acha.2024.101713

Personal Information

Date of Birth: 10 January 1998

Citizenship: USA

Last Updated: July 19, 2026